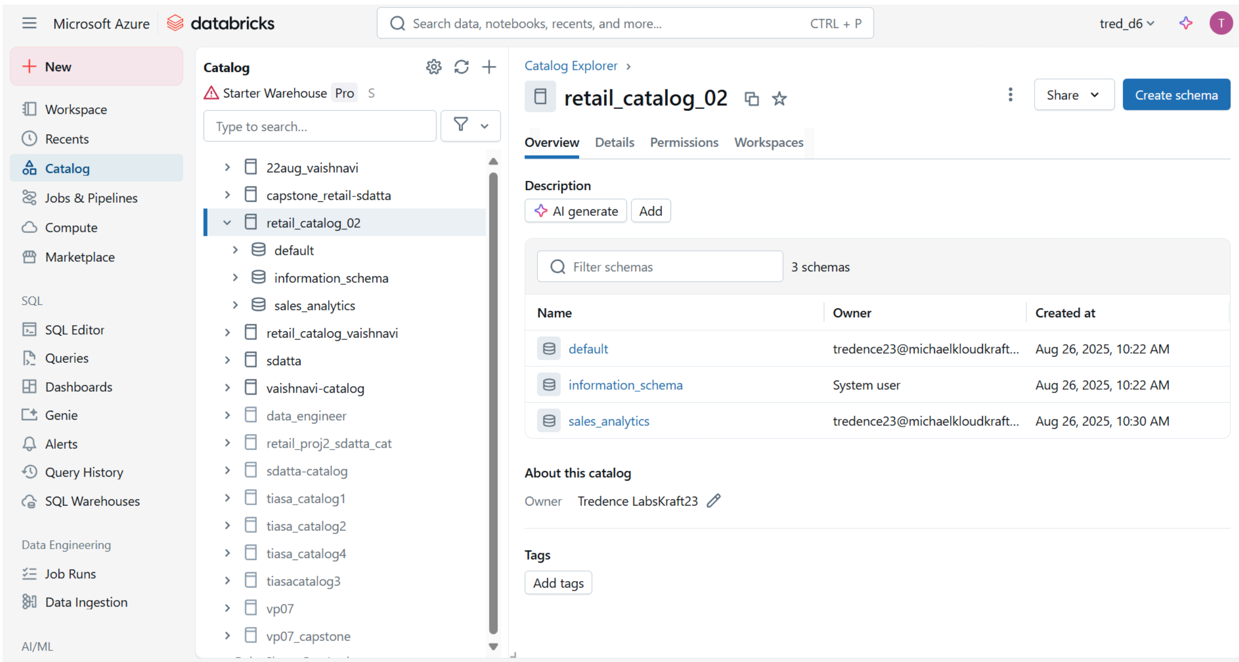
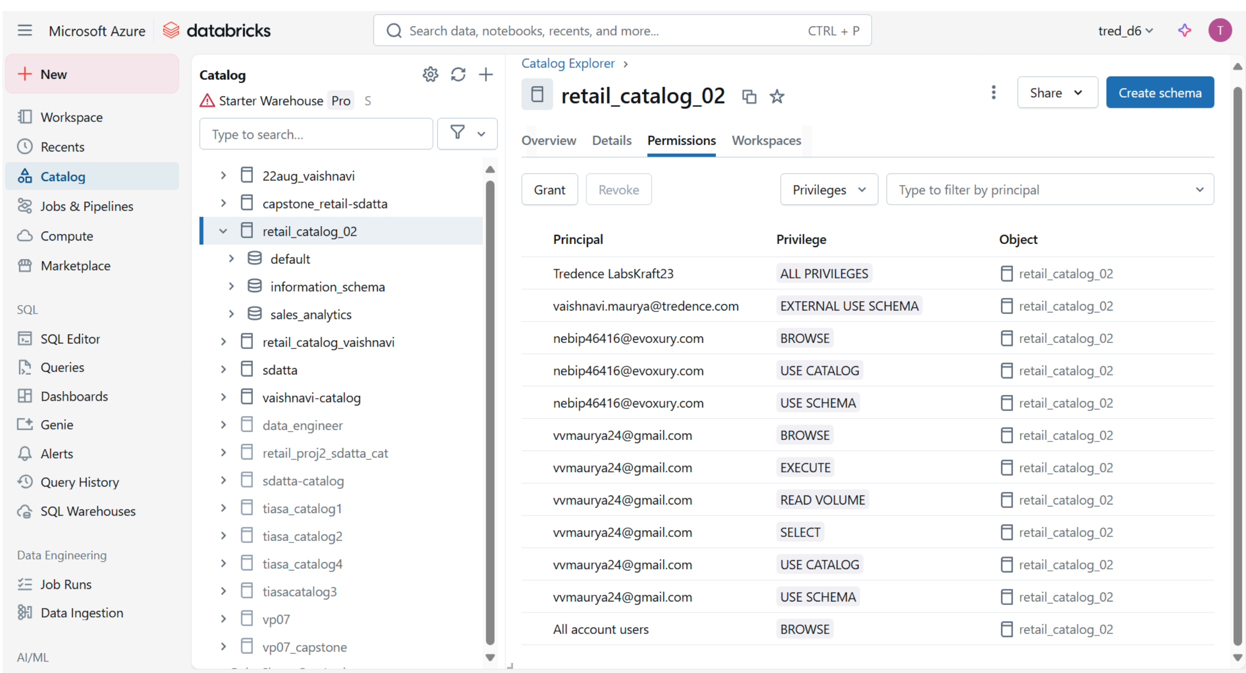
Step-1: Create storage account .

Step-2: Create a catalog- retail_catalog_02
Schema -> sales_analytics



Step-3: Grant required permission to the catalog.



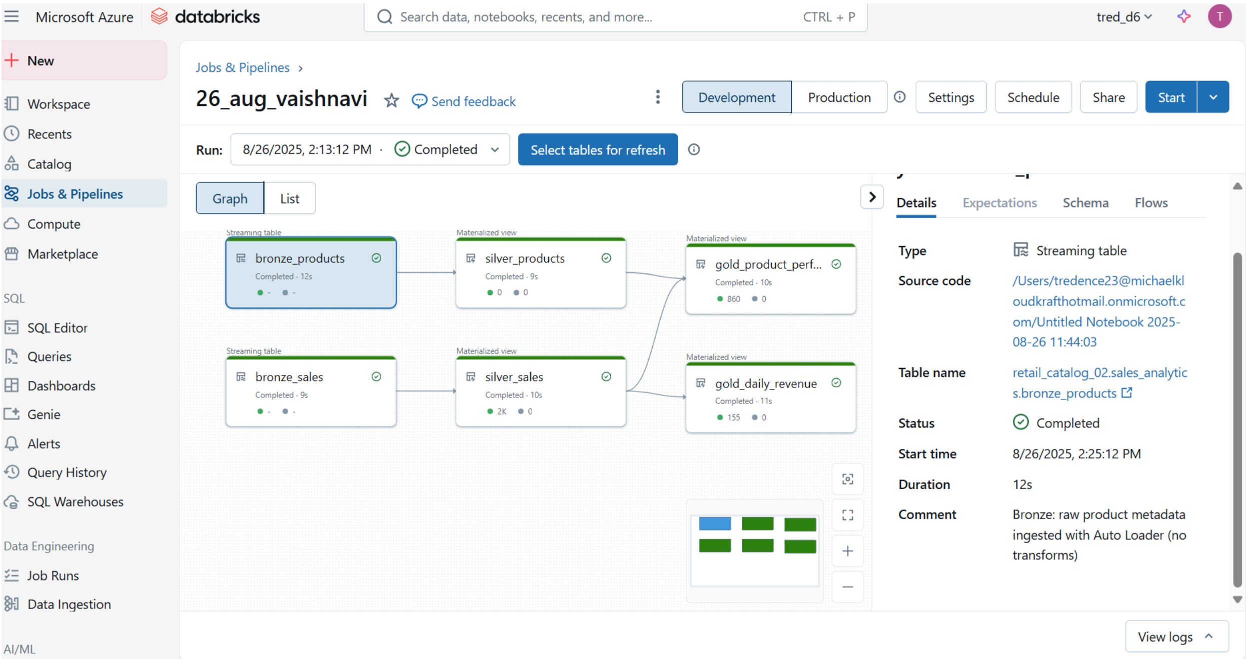
Step-4: Ingest data into bronze->silver->gold.

```
# retail_sales_pipeline.py
import dlt
from pyspark.sql.functions import *

from pyspark.sql.types import *
sales_schema = StructType([
    StructField("sale_id", IntegerType(), True),
    StructField("product_id", IntegerType(), True),
    StructField("region", StringType(), True),
    StructField("store_id", StringType(), True),
    StructField("quantity", IntegerType(), True),
    StructField("sales_amount", DoubleType(), True),
    StructField("sale_date", TimestampType(), True)
])
products_schema = StructType([
    StructField("product_id", StringType(), True),
    StructField("product_name", StringType(), True),
    StructField("category", StringType(), True),
    StructField("price", DoubleType(), True),
    StructField("discount_rate", DoubleType(), True)
])
# -------------------- CONFIG --------------------
RAW_SALES_PATH    = "abfss://raw@vaishnavidata.dfs.core.windows.net/Sales/"
RAW_PRODUCTS_PATH = "abfss://raw@vaishnavidata.dfs.core.windows.net/Products/"

# Schema/history locations (must be writable by the pipeline cluster)
SCHEMA_LOCATION_SALES    = "/dbfs/dlt_schema/retail/sales"    # DBFS path
example
SCHEMA_LOCATION_PRODUCTS = "/dbfs/dlt_schema/retail/products" # DBFS path
example
```

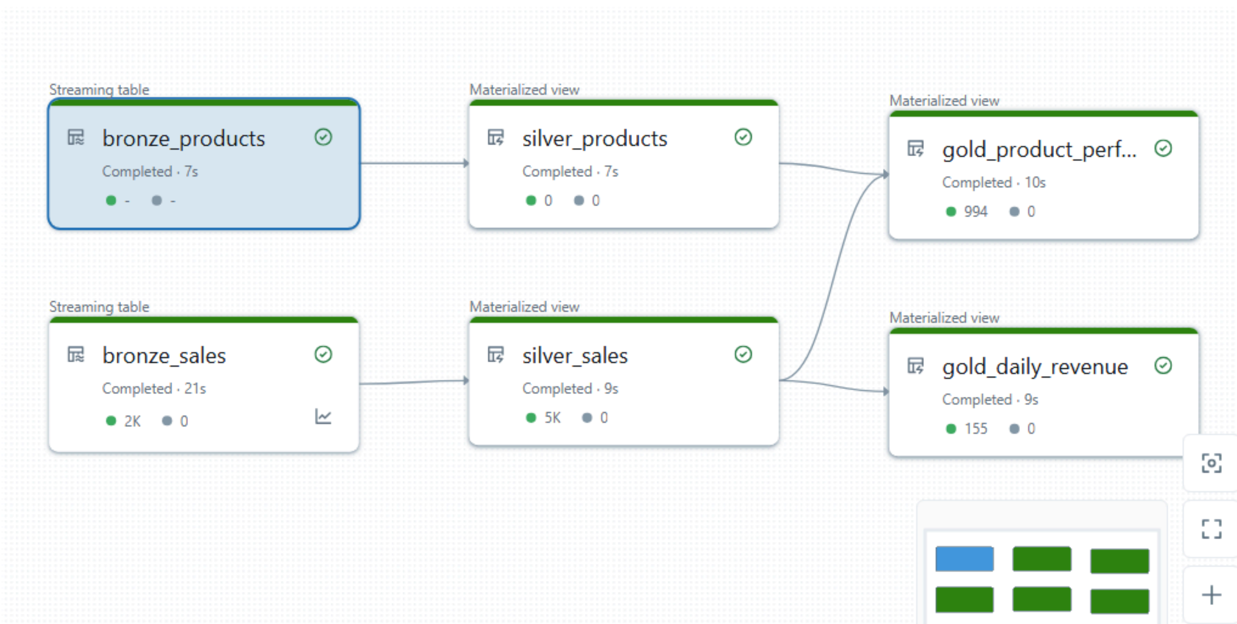
```python
# Target Unity Catalog (catalog.schema) for tables
TARGET_SCHEMA = "retail_catalog_02.sales_analytics"

# Helper to produce full UC table names
def tbl(name):
    return f"{TARGET_SCHEMA}.{name}"

# -------------------- BRONZE (raw ingestion) --------------------
@dlt.table(
    name=tbl("bronze_sales"),
    comment="Bronze: raw sales ingested with Auto Loader (no transforms)",
    table_properties={"quality": "bronze"}
)
def bronze_sales():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "csv")
            .option("header", "true")
            .schema(sales_schema)
            # .option("inferSchema", "true")                          # avoid
relying on inference here
            .option("cloudFiles.schemaLocation", SCHEMA_LOCATION_SALES)
            # .option("cloudFiles.schemaEvolutionMode", "addNewColumns")  #
allow new columns to appear
            .load(RAW_SALES_PATH)
    )

@dlt.table(
    name=tbl("bronze_products"),
    comment="Bronze: raw product metadata ingested with Auto Loader (no
transforms)",
    table_properties={"quality": "bronze"}
)
def bronze_products():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "json")   # change to 'csv' if your
product files are CSV
```

```
            .option("header", "true")
            .schema(products_schema)
            .option("cloudFiles.schemaLocation", SCHEMA_LOCATION_PRODUCTS)
            # .option("cloudFiles.schemaEvolutionMode", "addNewColumns")
            .load(RAW_PRODUCTS_PATH)
    )
@dlt.table(
    name="silver_sales",
    comment="Cleaned sales data (Silver)"
)
@dlt.expect_or_drop("non_null_orderid", "sale_id IS NOT NULL")
@dlt.expect("valid_quantity", "quantity >= 0")
def silver_sales():
    return (
        dlt.read("bronze_sales")
        .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
        .filter(col("sale_id").isNotNull())
    )

@dlt.table(
    name="silver_products",
    comment="Cleaned products data (Silver)"
)
@dlt.expect_or_drop("non_null_productid", "product_id IS NOT NULL")
def silver_products():
    return (
        dlt.read("bronze_products")
        .filter(col("product_id").isNotNull())
    )

# ======================================================
# 🔹 GOLD (Business Aggregates)
# ======================================================
@dlt.table(
    name="gold_daily_revenue",
    comment="Daily revenue per region (Gold)"
)
def gold_daily_revenue():
```

```
    return (
        dlt.read("silver_sales")
        .groupBy("region", "sale_date")
        .agg(
            sum(col("sales_amount")).alias("daily_revenue")  # ✅ clean alias
        )
    )

@dlt.table(
    name="gold_product_performance",
    comment="Aggregated product performance (Gold)"
)
def gold_product_performance():
    return (
        dlt.read("silver_sales")
        .join(dlt.read("silver_products"), on="product_id", how="left")
        .groupBy("product_id", "product_name", "category")
        .agg(
            sum("quantity").alias("total_units_sold"),
            count("sale_id").alias("total_orders")
        )
    )
```



After adding remaining files to container the records increase.



Added this file as record no. 5001 with extra columns without adding new columns.

A	B	C	D	E	F	G	H	I	J
sale_id	product_id	region	store_id	quantity	sales_amount	sale_date	Promotio_code	hello_code	fivetrancode
1	739	US	Store_12	14	150.45	8/12/2025 15:52	kesh40	90	uy6_poskjbksbv

	sale_id	product_id	region	store_id	quantity	sales_amount	sale_date
4992	992	532	EU	Store_43	16	1036.55	2025-08-12T22:38:
4993	993	31	EU	Store_37	10	1634.36	2025-08-08T15:20:
4994	994	60	MEA	Store_4	18	309.23	2025-08-03T22:31:
4995	995	100	EU	Store_5	12	1700.15	2025-08-12T08:43:
4996	996	113	US	Store_25	9	210.93	2025-08-04T19:32:
4997	997	11	LATAM	Store_22	12	2160.63	2025-08-23T21:19:
4998	998	170	LATAM	Store_25	7	631.87	2025-08-20T03:37:
4999	999	921	EU	Store_20	12	153.28	2025-08-22T20:09:
5000	1000	719	APAC	Store_44	19	3138.18	2025-08-25T20:05:
5001	1	739	US	Store_12	14	150.45	null
5002	null	null	null	null	null	null	null
5003	null	null	null	null	null	null	null

Pipeline event log details

This event is associated with `retail_catalog_02.sales_analytics.bronze_sales`.

Summary JSON

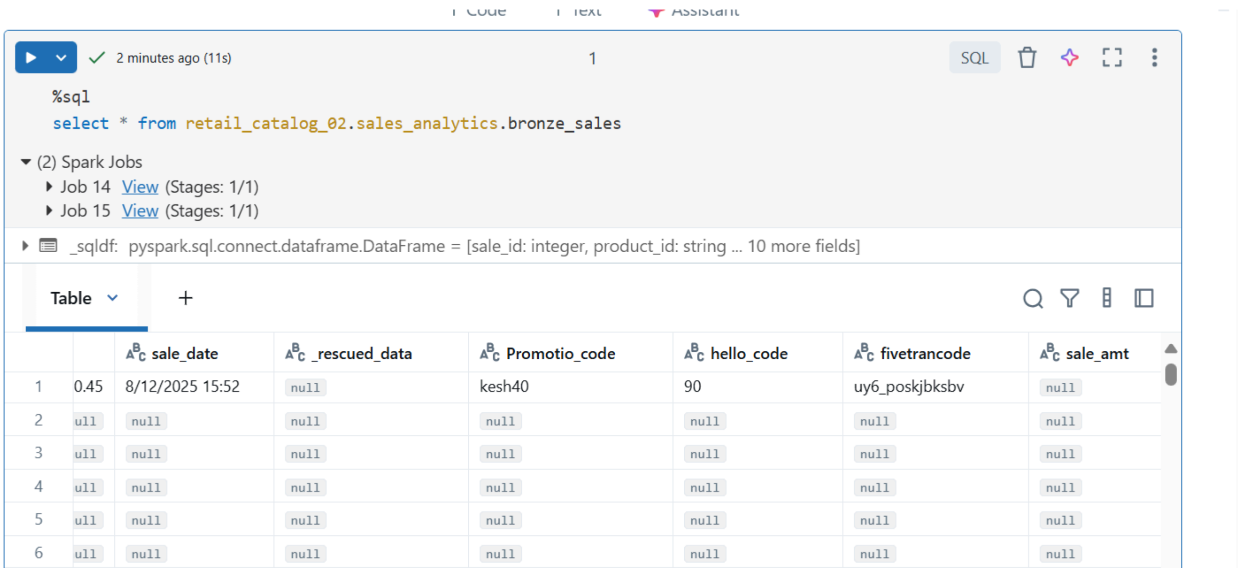
ERROR 2025-08-26 15:57:59 IST flow_progress

Failed to resolve flow: 'retail_catalog_02.sales_analytics.bronze_sales'. Diagnose error

Error details

```
py4j.protocol.Py4JJavaError: Traceback (most recent call last):
py4j.protocol.Py4JJavaError: An error occurred while calling o953.load.
: com.databricks.sql.cloudfiles.errors.CloudFilesIllegalArgumentException: Schema evolution mode addNewColumns is not supported when the schema is specified. To use this mode, you can provide the schema through `cloudFiles.schemaHints` instead.
        at com.databricks.sql.cloudfiles.errors.CloudFilesErrors$.addNewNotSupported(CloudFilesErrors.scala:115)
        at com.databricks.sql.cloudfiles.options.CloudFilesOptionsBase.assertSchemaEvolutionModeSupported(CloudFilesOptionsBase.scala:335)
        at com.databricks.sql.cloudfiles.options.CloudFilesOptionsBase.<init>(CloudFilesOptionsBase.scala:195)
        at com.databricks.sql.fileNotification.autoIngest.CloudFilesSourceOptions.<init>(CloudFilesSourceOptions.scala:54)
        at com.databricks.sql.fileNotification.autoIngest.CloudFilesSourceProvider.sourceSchema(CloudFilesSourceProvider.scala:84)
        at org.apache.spark.sql.execution.datasources.DataSource.sourceSchema(DataSource.scala:274)
        at org.apache.spark.sql.execution.datasources.DataSource.sourceInfo$lzycompute(DataSource.scala:158)
```

I hardcoded the schema so on schema evolution as add new column so error is coming.



Extra columns added after adding:

```
  .option("cloudFiles.schemaEvolutionMode", "addNewColumns")
            .option("cloudFiles.schemaHints", "sale_id INT, region STRING, 
sales_amount DOUBLE")
```

To the def bronze_sales()

Versioning

Just now (5s)

```sql
%sql
DESCRIBE HISTORY retail_catalog_02.sales_analytics.bronze_sales;
```

(1) Spark Jobs

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [version: long, timestamp: timestamp ... 13 more fields]

Table

	version	timestamp	userId	userName	operation
1	17	2025-08-26T10:54:43.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	STREAMING UP
2	16	2025-08-26T10:54:37.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	STREAMING UP
3	15	2025-08-26T10:53:51.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	DLT SETUP
4	14	2025-08-26T10:50:42.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	STREAMING UP
5	13	2025-08-26T10:49:49.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	DLT SETUP
6	12	2025-08-26T10:48:39.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	DLT REFRESH
7	11	2025-08-26T10:18:42.000+00:00	144583653157838	null	VACUUM END
8	10	2025-08-26T10:18:37.000+00:00	144583653157838	null	VACUUM START
9	9	2025-08-26T10:17:42.000+00:00	144583653157838	null	OPTIMIZE
10	8	2025-08-26T10:13:52.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	STREAMING UP
11	7	2025-08-26T10:04:54.000+00:00	144583653157838	tredence23@michaelkloudkrafthotmail.onmicrosoft.co...	STREAMING UP

```sql
%sql
SELECT *
FROM retail_catalog_02.sales_analytics.bronze_sales VERSION AS OF 6;
```

Table

	sale_id	product_id	region	store_id	quantity	sales_amount	sale_date
1	1	737	APAC	Store_40	16	655.38	2025-07-31T01:49:
2	2	185	LATAM	Store_48	6	872.66	2025-08-17T10:44:
3	3	18	EU	Store_20	11	317.64	2025-08-19T11:43:
4	4	744	EU	Store_21	2	276.58	2025-08-19T19:01:
5	5	878	US	Store_44	4	250.87	2025-08-14T19:25:
6	6	436	MEA	Store_13	2	139.47	2025-08-07T17:03:
7	7	69	APAC	Store_48	10	384.03	2025-08-12T16:09:
8	8	160	APAC	Store_20	4	477.14	2025-08-08T05:08:
9	9	62	EU	Store_30	1	25.4	2025-08-20T15:21:
10	10	942	EU	Store_22	18	2252.79	2025-07-31T02:22:
11	11	189	MEA	Store_42	18	2905.3	2025-08-03T22:43:
12	12	491	MEA	Store_35	18	3106.7	2025-08-08T11:39:
13	13	193	MEA	Store_21	17	3313.8	2025-08-08T21:27:
14	14	177	MEA	Store_29	12	107.04	2025-08-21T00:03:
15							

6,000 rows | 11.98s runtime

Refreshed now

The no. of rows reduced to 6000 from 8000.

Not able to use powerbi